

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
dataset=sns.load_dataset('titanic')
```

```
dataset.head()
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked
class \								
0 Third	0	3	male	22.0	1	0	7.2500	S
1 First	1	1	female	38.0	1	0	71.2833	C
2 Third	1	3	female	26.0	0	0	7.9250	S
3 First	1	1	female	35.0	1	0	53.1000	S
4 Third	0	3	male	35.0	0	0	8.0500	S

	who	adult_male	deck	embark_town	alive	alone
0	man	True	NaN	Southampton	no	False
1	woman	False	C	Cherbourg	yes	False
2	woman	False	NaN	Southampton	yes	True
3	woman	False	C	Southampton	yes	False
4	man	True	NaN	Southampton	no	True

```
dataset.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 15 columns):
#   Column                Non-Null Count  Dtype
---  -
0   survived              891 non-null    int64
1   pclass                891 non-null    int64
2   sex                  891 non-null    object
3   age                  714 non-null    float64
4   sibsp                891 non-null    int64
5   parch                891 non-null    int64
6   fare                 891 non-null    float64
7   embarked             889 non-null    object
8   class                891 non-null    category
9   who                  891 non-null    object
10  adult_male           891 non-null    bool
11  deck                 203 non-null    category
12  embark_town          889 non-null    object
13  alive                891 non-null    object
```

```
14 alone      891 non-null    bool
dtypes: bool(2), category(2), float64(2), int64(4), object(5)
memory usage: 80.7+ KB
```

```
dataset.describe()
```

	survived	pclass	age	sibsp	parch
fare					
count	891.000000	891.000000	714.000000	891.000000	891.000000
891.000000					
mean	0.383838	2.308642	29.699118	0.523008	0.381594
32.204208					
std	0.486592	0.836071	14.526497	1.102743	0.806057
49.693429					
min	0.000000	1.000000	0.420000	0.000000	0.000000
0.000000					
25%	0.000000	2.000000	20.125000	0.000000	0.000000
7.910400					
50%	0.000000	3.000000	28.000000	0.000000	0.000000
14.454200					
75%	1.000000	3.000000	38.000000	1.000000	0.000000
31.000000					
max	1.000000	3.000000	80.000000	8.000000	6.000000
512.329200					

```
dataset.isNull()
```

[illegible]

False

	who	adult_male	deck	embark_town	alive	alone
0	False	False	True	False	False	False
1	False	False	False	False	False	False
2	False	False	True	False	False	False
3	False	False	False	False	False	False
4	False	False	True	False	False	False
...	...	...	...	...	...	...
886	False	False	True	False	False	False
887	False	False	False	False	False	False
888	False	False	True	False	False	False
889	False	False	False	False	False	False
890	False	False	True	False	False	False

[891 rows x 15 columns]

```
dataset.isnull().sum()
```

survived	0
pclass	0
sex	0
age	177
sibsp	0
parch	0
fare	0
embarked	2
class	0
who	0
adult_male	0
deck	688
embark_town	2
alive	0
alone	0

dtype: int64

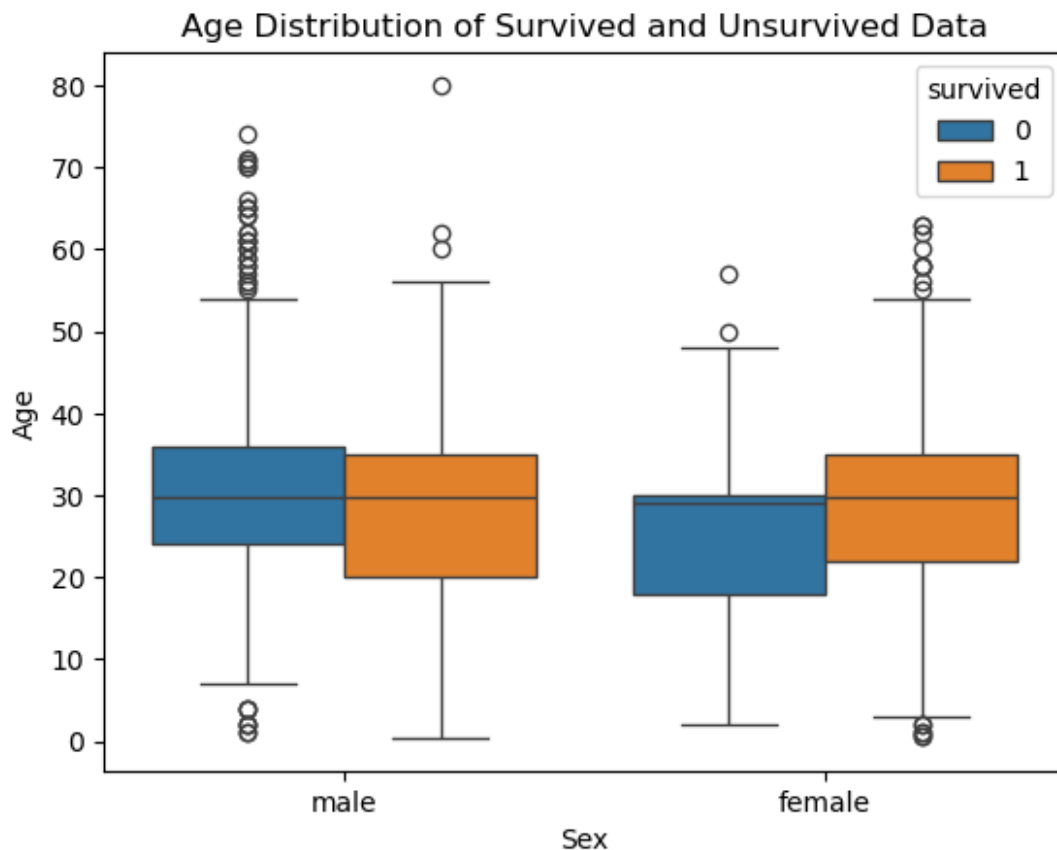
```
dataset['age'] = dataset['age'].fillna(dataset['age'].mean())
```

```
dataset.isnull().sum()
```

survived	0
pclass	0
sex	0
age	0
sibsp	0
parch	0
fare	0
embarked	2
class	0
who	0
adult_male	0

```
deck          688
embark_town    2
alive         0
alone         0
dtype: int64
```

```
sns.boxplot(x='sex',y='age',hue = 'survived',data=dataset)
plt.title('Age Distribution of Survived and Unsurvived Data')
plt.xlabel('Sex')
plt.ylabel('Age')
plt.show()
```



Female passengers show a noticeably higher survival rate compared to male passengers. Male non-survivors dominate the middle-age range, while male survivors are generally younger. Children (low age values) have better survival chances across both sexes. The median age of male survivors is lower than that of male non-survivors. Female survivors display a wider age spread compared to female non-survivors. Outliers include very old passengers in both survived and non-survived groups. The interquartile range for female survivors is larger than for female non-survivors. Overall, survival patterns differ significantly between male and female age distributions.